

Plan: Extend cq-level Range from [0, 63] to [0, 80]

Context

In AOM_Q (constant quality) mode, the user-facing cq-level parameter (0-63) maps to internal qindex values (0-255) via a lookup table. Due to quality boost multipliers applied per hierarchical layer (KF/ARF via `qstep_ratio`, intermediate ARF via midpoint interpolation), `cq-level=63` only produces qindex ~173 for key frames. The full qindex range [0, 255] is unreachable for boosted frame types.

Extending cq-level to 80 allows all frame types -- including key frames -- to reach qindex 255, giving the encoder full quantizer coverage across all hierarchical layers.

This feature is only needed in constant quality mode (`end-usage=q`). Bitrate control modes (CBR, VBR) are unaffected.

Scope

The extended cq-level range (64-80) is supported only in constant quality mode (AOM_Q) with `use-fixed-qp-offsets=1` and TPL disabled (`enable-tpl-model=0`):

- AOM_Q + `use-fixed-qp-offsets=1` + `enable-tpl-model=0`: Uses `qstep_ratio` approach in `encoder_utils.c` -- covered in Part 1
- All other modes/settings: `cq-level > 63` is rejected with an assertion and error message -- covered in Part 2

Part 1: Fixed QP Offsets Path (`use-fixed-qp-offsets=1`)

Approach: Virtual qindex > 255 with Linear `qstep_ratio` Ramp

Store unclamped "virtual" qindex values (259-323) in `oxcf->cq_level` for cq-levels 64-80.

For the `use_fixed_qp_offsets=1` code path in `encoder_utils.c`, a unified `qstep_ratio` approach is used across the entire cq-level range:

- cq 0-63 (qindex 0-255): `qstep_ratio = 0.2 + (1.0 - awq/255) * qratio_grad` -- ratio decreases from 0.5 to 0.2 as quality decreases (existing formula, unchanged)
- cq 64-80 (virtual qindex 259-323): `qstep_ratio = 0.2 + (cq_level - 255) / (323 - 255) * 0.8` -- ratio increases from 0.2 toward 1.0, so KF/ARF qindex converges to MAXQ

At `cq=63`, the ratio reaches 0.2 (minimum boost). For `cq 64-80`, the ratio smoothly ramps back toward 1.0, meaning KF/ARF qindex approaches the leaf qindex (255). At `cq=80`, all layers converge to qindex 255.

When `qstep_ratio >= 1.0` (which occurs at `cq=80`), the `av1_get_q_index_from_qstep_ratio()` search is bypassed and `min(cq_level, MAXQ)` is returned directly, ensuring exact convergence to qindex 255 without off-by-1 rounding from the DC quantizer lookup.

Verified qindex mapping (8-bit, `use-fixed-qp-offsets=1`)

Values verified against actual encoder output (FourPeople 480x270, 130 frames, `cpu-used=4`, 8-bit). KF and ARF produce identical qindex in fixed QP mode. At `cq <= 79`, KF/ARF shows off-by-1 from predicted values due to DC quantizer rounding in `av1_get_q_index_from_qstep_ratio()`; IARF and Leaf are exact. At `cq=80`, the `qstep_ratio >= 1.0` shortcut ensures all layers reach exactly 255.

cq	qindex	KF/ARF	IARF_d2	IARF_d3	IARF_d4	Leaf
0	0	0	0	0	0	0
10	40	13	27	34	37	40
20	80	27	54	67	74	80
30	120	44	82	101	111	120

40	160	74	117	139	150	160
50	200	106	153	177	189	200
55	220	121	171	196	208	220
60	240	142	191	216	228	240
63	255	172	214	235	245	255
64	259	188	222	239	247	255
66	267	211	233	244	250	255
68	275	226	241	248	252	255
70	283	235	245	250	253	255
72	291	241	248	252	254	255
74	299	246	251	253	254	255
76	307	249	252	254	255	255
78	315	252	254	255	255	255
79	319	253	254	255	255	255
80	323	255	255	255	255	255

cq-level to qindex plot (fixed QP)

[Image: cq-level to qindex mapping -- use-fixed-qp-offsets=1]

Changes for Fixed QP Path

1. Extend quantizer_to_qindex table and reverse function

File: av1/encoder/av1_quantize.c

- Update comment: "0-63" to "0-80"
- Add 17 entries continuing from 255 with step +4 (unclamped virtual values):
259, 263, 267, 271, 275, 279, 283, 287, 291,
295, 299, 303, 307, 311, 315, 319, 323,
- av1_qindex_to_quantizer(): change loop bound < 64 to < 81, fallback return 63 to return 80
- Total table size: 81 entries (cq 0-80)

2. Clamp best_allowed_q and worst_allowed_q

File: av1/av1_cx_iface.c

best_allowed_q and worst_allowed_q feed directly into quantization and must stay in [0, 255]. cq_level is only used in rate control delta math and can exceed 255:

```
oxcf->best_allowed_q =
    extra_cfg->lossless ? 0 : AOMMIN(av1_quantizer_to_qindex(cfg->rc_min_quantizer), MAXQ);
oxcf->worst_allowed_q =
    extra_cfg->lossless ? 0 : AOMMIN(av1_quantizer_to_qindex(cfg->rc_max_quantizer), MAXQ);
oxcf->cq_level = av1_quantizer_to_qindex(extra_cfg->cq_level); // can be > 255
```

3. Guard av1_convert_qindex_to_q() against qindex > 255

File: av1/encoder/ratectrl.c

Add clamp at the top of the function so ac_qlookup[] is not accessed out of bounds:

```
double av1_convert_qindex_to_q(int qindex, aom_bit_depth_t bit_depth) {
    qindex = AOMMIN(qindex, MAXQ); // clamp to valid ac_qlookup range
    // ... rest unchanged
}
```

4. Update range checks

File: av1/av1_cx_iface.c

Location	Current	New	Notes
RANGE_CHECK_HI(extra_cfg, cq_level, 63)		80	Allow extended cq-level

Add conditional assertion after the range check -- reject cq > 63 unless AOM_Q + fixed QP + TPL disabled:

```
if (extra_cfg->cq_level > 63) {
    if (cfg->rc_end_usage != AOM_Q) {
        ERROR("cq-level > 63 is only supported in AOM_Q (constant quality) mode");
    }
}
```

```

    }
    if (!extra_cfg->use_fixed_qp_offsets) {
        ERROR("cq-level > 63 requires use-fixed-qp-offsets=1");
    }
    if (extra_cfg->enable_tpl_model) {
        ERROR("cq-level > 63 requires enable-tpl-model=0");
    }
}

```

Not changed:

- RANGE_CHECK_HI(cfg, rc_max_quantizer, 63) -- stays at 63 (used by CBR/VBR)
- 63, // rc_max_quantizer default -- stays at 63

5. Update fixed QP offsets formula for extended range

File: av1/encoder/encoder_utils.c

The existing code already has the extended range branch. Update max_virtual_qindex from 327 to 323, and add a qstep_ratio >= 1.0 shortcut to bypass av1_get_q_index_from_qstep_ratio() when the ratio means no boost:

```

if (cq_level <= MAXQ) {
    qstep_ratio = 0.2 + (1.0 - (double)cpi->rc.active_worst_quality / MAXQ) * qratio_grad;
} else {
    const int max_virtual_qindex = 323; // quantizer_to_qindex[80]
    qstep_ratio = 0.2 + (double)(cq_level - MAXQ) / (max_virtual_qindex - MAXQ) * 0.8;
}
if (qstep_ratio >= 1.0)
    *q = AOMMIN(cq_level, MAXQ);
else
    *q = av1_get_q_index_from_qstep_ratio(
        cpi->rc.active_worst_quality, qstep_ratio,
        cm->seq_params->bit_depth);

```

Why the shortcut is needed: av1_get_q_index_from_qstep_ratio() searches downward from leaf_qindex for a qindex whose DC qstep satisfies $qstep + 0.1 \leq target_qstep$. When $qstep_ratio = 1.0$, $target_qstep = leaf_qstep$, so $leaf_qstep + 0.1 \leq leaf_qstep$ is false and the search drops to $leaf_qindex - 1$ (254 instead of 255). The shortcut avoids this off-by-1 at the boundary, ensuring all layers converge to exactly 255 at $cq=80$.

The shortcut only activates at $cq=80$ where the ratio reaches exactly 1.0. For all other cq levels (0-79), the ratio is < 1.0 and the normal search path is used.

6. No changes needed

- aomenc.c: CLI option definitions have no hardcoded range -- validation is in av1_cx_iface.c
- warnings.c: check_quantizer() only warns about narrow Q range
- QINDEX_RANGE, MAXQ, ac_qlookup[]: Internal qindex space [0, 255] is unchanged
- Rate control minq tables: Indexed by qindex [0, 255], unchanged
- Existing clamp() calls in rc_pick_q_and_bounds_: Already clamp final q to [rc->best_quality, rc->worst_quality] = [0, 255]

Part 2: Guard for Unsupported Modes

Approach: Assert and Reject

For any configuration other than AOM_Q + use-fixed-qp-offsets=1 + enable-tpl-model=0, cq-level values above 63 are not supported. The encoder validates this at configuration time and rejects invalid combinations with a clear error message.

The validation logic is added in av1_cx_iface.c (see Part 1, item 4 above).

Rationale

The non-fixed QP path (use-fixed-qp-offsets=0) uses polynomial minq table lookups and av1_compute_qdelta with content-adaptive state (avg_frame_qindex). Extending virtual qindex values into this path introduces

inconsistencies -- for example, the first ARF frame gets qindex=0 due to boost-table lookups (get_gf_active_quality()) that ignore cq_level and rely on avg_frame_qindex, which is near-zero early in the sequence. Rather than patching multiple rate control code paths with workarounds, the extended range is restricted to the deterministic fixed-QP path where qindex assignment is fully controlled by the qstep_ratio formula.

When TPL is enabled (enable-tpl-model=1), the temporal propagation model can override the fixed QP offsets, making the qstep_ratio-based qindex assignment non-deterministic. The extended cq-level range relies on the deterministic fixed-QP path, so TPL must be disabled.

Similarly, CQ/CBR/VBR modes either don't use cq_level or use it only for comparisons that assume qindex <= 255. Extending into those modes offers no benefit and risks subtle breakage.

Safety Analysis: Impact on Other Rate Control Modes

Mode	Uses cq_level?	Impact	Status
AOM_CBR	No	Completely unaffected	SAFE
AOM_VBR	No	Completely unaffected	SAFE
AOM_CQ	Yes (comparisons)	cq_level > 63 rejected at config validation	BARDED
AOM_Q (fixed QP)	Yes (direct)	Extended range handled by qstep_ratio with shortcut	SAFE
AOM_Q (non-fixed)	Yes (direct)	cq_level > 63 rejected at config validation	BARDED

Key safety guarantees:

- av1_ac_quant() in quant_common.c already has internal clamp(qindex + delta, 0, MAXQ) -- defense-in-depth
- All rc_pick_q_and_bounds_ functions clamp final q to [rc->best_quality, rc->worst_quality] = [0, 255]
- best_allowed_q/worst_allowed_q flow into rc->worst_quality which is used as array index for inter_minq[]/rtc_minq[] (256 elements) -- must stay <= 255 (Fix #2)
- cq_level is never used as a direct array index -- only in arithmetic and as input to av1_convert_qindex_to_q() (Fix #3)
- The qstep_ratio >= 1.0 shortcut in encoder_utils.c only activates at cq=80 and returns AOMMIN(cq_level, MAXQ) = 255
- Config validation rejects cq > 63 for any mode other than AOM_Q + use-fixed-qp-offsets=1 + enable-tpl-model=0

Files Modified

File	Changes	Status
av1/encoder/av1_quantize.c	Extend table to 81 entries; update reverse lookup	Done
av1/av1_cx_iface.c	cq_level range 63->80; clamp best/worst_allowed_q	Done
av1/encoder/ratectrl.c	Clamp qindex in av1_convert_qindex_to_q()	Done
av1/encoder/encoder_utils.c	Update max_virtual_qindex 327->323; add qstep_ratio >= 1.0 shortcut	Done

Test Configuration

The primary test configuration uses use_fixed_qp_offsets=1 with deltaq-mode=0 to achieve deterministic, content-independent qindex assignment per hierarchical layer.

Encoding Command

```
aomenc --verbose --codec=av1 -v --psnr --obu \
--frame-parallel=0 --cpu-used=0 --limit=600 --passes=1 --end-usage=q \
--i420 --use-fixed-qp-offsets=1 --deltaq-mode=0 --enable-tpl-model=0 \
--fps=60/1 --input-bit-depth=10 --bit-depth=10 --cq-level=$CQ \
--tile-columns=0 --threads=1 --enable-keyframe-filtering=1 \
--kf-min-dist=65 --kf-max-dist=65 --min-gf-interval=16 --max-gf-interval=16 \
--gf-min-pyr-height=4 --gf-max-pyr-height=4 --lag-in-frames=19 --auto-alt-ref=1 \
-w 1920 -h 1080 \
```

```
-o output.obu input.y4m
```

Key flags explained

Flag	Value	Purpose
--use-fixed-qp-offsets=1	1	Bypass polynomial minq lookup tables; use unified qstep_ratio a
--deltaq-mode=0	0	Disable per-SB delta-q adaptation; uniform qindex within each fr
--enable-tpl-model=0	0	Disable temporal propagation model; prevent TPL from overridin
--passes=1	1	Single-pass encode; uses rc_pick_q_and_bounds_no_stats for r
--kf-min-dist=65 --kf-max-dist=65	65	Fixed keyframe interval at every 65 frames
--min-gf-interval=16 --max-gf-interval=16	16	Fixed golden frame interval at 16 frames
--gf-min-pyr-height=4 --gf-max-pyr-height=4	4	4-level hierarchical pyramid (ARF -> IARF_d2 -> IARF_d3 -> IAR
--bit-depth=10	10	10-bit encoding; uses dc_qlookup_10_QTX[] for quantizer looku

Q selection code path with this configuration

With use_fixed_qp_offsets=1, the final qindex override happens in av1_set_size_dependent_vars() (encoder_utils.c), which runs after the initial av1_rc_pick_q_and_bounds() selection:

- KF/ARF/GF frames (is_frame_tpl_eligible() returns true):
 - cq_level <= 255: qstep_ratio = 0.2 + (1.0 - awq/255) * qratio_grad, search downward for matching qindex via av1_get_q_index_from_qstep_ratio()
 - cq_level > 255 (cq 64-80): qstep_ratio = 0.2 + (cq_level - 255) / (323 - 255) * 0.8, same search with leaf_qindex = min(cq_level, 255) = 255
 - When qstep_ratio >= 1.0 (cq=80): shortcut returns min(cq_level, 255) = 255 directly
 - KF and ARF use the same qstep_ratio (unified, no separate boost factors)
- INTNL_ARF frames (intermediate pyramid levels): midpoint interpolation q = (arf_q + min(cq_level, 255) + 1) / 2 per depth level
- Leaf/Overlay frames: q = min(cq_level, 255)

This produces a deterministic qindex-per-layer mapping that depends only on cq_level.

Verification

Build

```
cmake path/to/aom && make -j$(nproc)
```

Fixed QP path (use-fixed-qp-offsets=1)

Test	Command	Expected Result	Status
Original range	--cq-level=32	No regression	Pass
Extended range	--cq-level=70	Valid output, per-layer qindex different	Pass
Max cq	--cq-level=80	All layers reach qindex 255	Pass
Out of range	--cq-level=81	Rejected with error	Pass

Bitrate (480x270, 130 frames, cpu-used=4):

cq	Bitrate (bps)	PSNR-Y (dB)
60	154,870	34.38
63	100,685	31.58
64	82,633	30.28
70	39,651	25.98
80	27,969	23.63

Unsupported mode rejection

Test	Command	Expected Result	Status
Non-fixed QP + extended cq	--end-usage=q --use-fixed-qp-offsets=1	Rejected: "cq-level > 63 requires use-fixed-qp-offsets=1"	Pass
TPL enabled + extended cq	--end-usage=q --use-fixed-qp-offsets=1 --enable-tpl-model=0	Rejected: "cq-level > 63 requires enable-tpl-model=0"	Pass
CQ mode + extended cq	--end-usage=cq --cq-level=70	Rejected: "cq-level > 63 is only supported in AOM_Q mode"	Pass

CBR + extended cq	--end-usage=cbr --cq-level=70	Rejected: "cq-level > 63 is only supported in AOM_Q mode"
VBR + extended cq	--end-usage=vbr --cq-level=70	Rejected: "cq-level > 63 is only supported in AOM_Q mode"
Non-fixed QP + original range	--end-usage=q --use-fixed-qp-offsets=10 --qp-level=32	--

Other modes

Test	Command	Expected Result	Status
CBR mode	--end-usage=cbr --target-bitrate=1000	No regression (cq_level not used)	Pass
CQ mode original range	--end-usage=cq --cq-level=50	Runs correctly	Pass
Unit tests	make runtests	All pass	--

Reference

- Extended range plot (fixed QP): cq_extended_fixed_maxcq80.svg
- Unified plot: qindex_per_layer_chart_unified.svg
- Original range comparison: cq_to_qindex_fixed.svg (fixed QP) / cq_to_qindex_nonfixed.svg (non-fixed QP)
- Alternative max_cq options: cq_extended_fixed_maxcq70.svg, cq_extended_fixed_maxcq75.svg
- Data: cq_to_qindex_data.csv
- Computation script: plot_cq_to_qindex.py